Edexcel GCSE Physics **Student Progress Sheet**

Name: _____ **Target:** _____

GCSE Physics

Topic 6 Radioactivity



6.1	I can describe an atom as a positively charged nucleus consisting of protons and neutrons. This is surrounded by negatively charged electrons. The nuclear radius is much smaller than that of the atom and with almost all of the mass in the nucleus.			
6.2	I can recall the typical size of atoms and small molecules.			
6.3	I can describe the structure of nuclei of isotopes using the terms atomic (proton) number and mass (nucleon) number, and using symbols in the format: ^{6} ^{13}C			
6.4	I can recall that the nucleus of each element has a characteristic positive charge, but that isotopes of an element differ in mass by having different numbers of neutrons.			
6.5	I can recall the relative masses and relative electric charges of protons, neutrons, electrons and positrons.			
6.6	I can recall that, in an atom, the number of protons equals the number of electrons and is therefore neutral.			
6.7	I can recall that, in each atom, the electrons orbit the nucleus at different set distances from the nucleus.			
6.8	I can explain that electrons change orbit when there is absorption or emission of electromagnetic radiation.			
6.9	I can explain how atoms may form positive ions by losing outer electrons.			
6.10	I can recall that alpha, β– (beta minus), β+ (positron), gamma rays and neutron radiation are emitted from unstable nuclei in a random process.			
6.11	I can recall that alpha, β– (beta minus), β+ (positron) and gamma rays are ionising radiations.			

6.12	I can explain what is meant by background radiation.			
6.13	I can describe the origins of background radiation from Earth and space.			
6.14	I can describe methods for measuring and detecting radioactivity limited to photographic film and a Geiger–Müller tube.			
6.15	I can recall that an alpha particle is equivalent to a helium nucleus, a beta particle is an electron emitted from the nucleus, and a gamma ray is electromagnetic radiation.			
6.16	I can compare alpha, beta and gamma radiations in terms of their abilities to penetrate and ionize.			
6.17	I can describe how and why the atomic model has changed over time, including reference to the plum pudding model and Rutherford alpha particle scattering, leading to the Bohr model.			
6.18	I can describe the process of $\beta-$ decay (a neutron becomes a proton plus an electron).			
6.19	I can describe the process of $\beta+$ decay (a proton becomes a neutron plus a positron).			
6.20	I can explain the effects on the atomic (proton) number and mass (nucleon) number of radioactive decays (α, β, γ and neutron emission).			
6.21	I can recall that nuclei that have undergone radioactive decay often undergo nuclear rearrangement with a loss of energy as gamma radiation.			
6.22	I can use given data to balance nuclear equations in terms of mass and charge.			
6.23	I can describe how the activity of a radioactive source decreases over a period of time.			
6.24	I can recall that the unit of activity of a radioactive isotope is the Becquerel, Bq.			
6.25	I can explain that the half-life of a radioactive isotope is the time taken for half the undecayed nuclei to decay or the activity of a source to decay by half.			
6.26	I can explain that it cannot be predicted when a particular nucleus will decay, but that half-life enables the activity of a very large number of nuclei to be predicted during the decaying process.			

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6.27	I can use the concept of half-life to carry out simple calculations on the decay of a radioactive isotope, including graphical representations.			
6.28	I can describe uses of radioactivity, including: a) household fire (smoke) alarms; b) irradiating food; c) sterilisation of equipment; d) tracing and gauging thicknesses; e) diagnosis and treatment of cancer.			
6.29	I can describe the dangers of ionising radiation in terms of tissue damage and possible mutations and relate this to the precautions needed.			
6.30	I can explain how the dangers of ionising radiation depend on half-life and relate this to the precautions needed.			
6.31	I can explain the precautions taken to ensure the safety of people exposed to radiation, including limiting the dose for patients and the risks to medical personnel.			
6.32	I can describe the differences between contamination and irradiation effects and compare the hazards associated with these two.			
6.33	I can compare and contrast the treatment of tumours using radiation applied internally or externally.			
6.34	I can explain some of the uses of radioactive substances in the diagnosis of medical conditions, including PET scanners and tracers.			
6.35	I can explain why isotopes used in PET scanners have to be produced nearby.			
6.36	I can evaluate the advantages and disadvantages of nuclear power for generating electricity, including the lack of carbon dioxide emissions, risks, public perception, waste disposal and safety issues.			
6.37	I can recall that nuclear reactions, including fission, fusion and radioactive decay, can be a source of energy.			
6.38	I can explain how the fission of U-235 produces two daughter nuclei and the emission of two or more neutrons, accompanied by a release of energy.			
6.39	I can explain the principle of a controlled nuclear chain reaction.			



6.40	I can explain how the chain reaction is controlled in a nuclear reactor, including the action of moderators and control rods.			
6.41	I can describe how thermal (heat) energy from the chain reaction is used in the generation of electricity in a nuclear power station.			
6.42	I can recall that the products of nuclear fission are radioactive.			
6.43	I can describe nuclear fusion as the creation of larger nuclei resulting in a loss of mass from smaller nuclei, accompanied by a release of energy, and recognise fusion as the energy source for stars.			
6.44	I can explain the difference between nuclear fusion and nuclear fission.			
6.45	I can explain why nuclear fusion does not happen at low temperatures and pressures, owing to electrostatic repulsion of protons.			
6.46	I can relate the conditions for fusion to the difficulty of making a practical and economic form of power station.			

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